

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 4 SET C

SUMMER SEMESTER, 2023-2024

DURATION: 25 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. A D flip-flop combined with some additional combinational logic is shown in Figure 1. Construct the truth table for the resulting flip-flop, analyzing all possible input combinations. Then, by using process of elimination, determine which standard flip-flop the circuit's behavior most closely resembles.

10
(CO5)
(PO3)

Hint: Consider each case individually and observe how the output changes in response to different inputs and previous states.

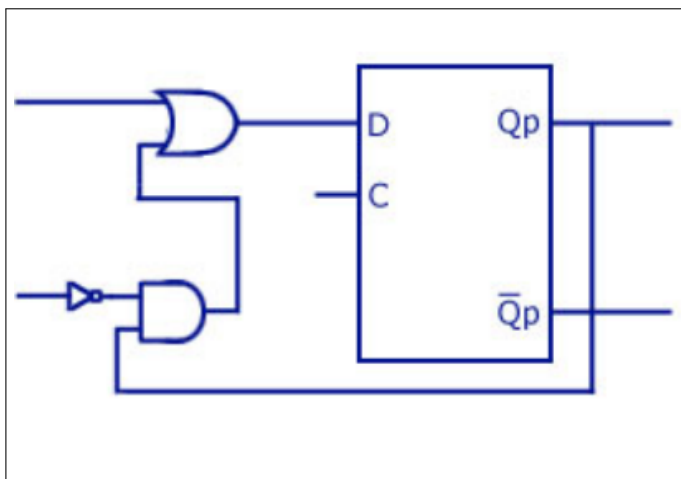


Figure 1: An unknown FlipFlop

2. Design a 5-bit Parallel-In Serial-Out (PISO) shift register (**shifting right to left**) with a control line labeled $\text{SHIFT} \setminus \overline{\text{LOAD}}$. **Explain the working procedure of the circuit in different modes.**

5 + 5
(CO5)
(PO3)

